

Employment

NC State University	Graduate Research Assistant	Apr 2025 – Present
- Collaborated with plant biologists and geneticists to deploy cloud-native genomic data platform on GCP (Cloud Run) with JBrowse 2 genome browser, enabling 100+ researchers to visualize 500GB+ genome datasets - Improved ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data		
Isomer AI	Product Engineer Intern	May 2025 – May 2026
- Shipped per-tenant AI workflow platform for insurance carriers: Temporal.io durable orchestration, LLM risk signal detection across Claude/GPT-4o/Gemini, typed document classification, Auth0 RBAC; 30+ deployments - Built MCP servers (Python + TypeScript); Playwright + Tailscale VPN for customer-internal browser automation		
Mphasis (HP Company)	Software Developer Intern	Jun 2024 – Aug 2024
- Partnered with healthcare data teams to build 5+ AI workflows including health claim automation using OCR, NLP, and ML pipelines; improved document processing accuracy by 25% across 10K+ daily medical claims - Automated AWS and GCP CI/CD pipelines; integrated Grafana monitoring and led model validation and red-teaming sessions		
Capgemini	AI/ML Intern	Jun 2023 – Aug 2023
- Led SDLC for enterprise ticket analysis using NLP classification models, cutting resolution time 15% and eliminating 2,000+ monthly manual triages - Built Kafka + Spark + Hadoop pipelines with data engineers; cut ticket-processing latency by 20% at peak load		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
Master of Science in Computer Science; GPA: 3.5/4.0. Courses: Artificial Intelligence, Machine Learning, Algorithms, Data Science		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Structures & Algorithms, Big Data Infrastructure, Database Systems		

Projects

VectorLift	Semantic Search & Ranking Engine	2026
- Built production search system on MS MARCO: BM25 baseline in Elasticsearch, a fine-tuned dense bi-encoder, and a cross-encoder re-ranker; served via FastAPI with per-stage latency tracking and a Streamlit comparison dashboard - Designed Kafka-driven indexing into Elasticsearch + Qdrant; evaluated 6 pipeline configs with NDCG@10/MRR@10/MAP and paired bootstrap significance so every ranking change stays defensible, not anecdotal		
InventoryReservationService	Transactional Java API	2026
- Engineered Java 21 + Spring Boot reservation API preventing overselling via PostgreSQL row-level locking (SELECT FOR UPDATE); idempotent endpoints with RFC 9457 errors, Flyway migrations, and Spring Security RBAC - Proved concurrency correctness with Testcontainers (two concurrent reservations on qty 1 yield exactly one success and one 409 Conflict); multi-stage Docker build and Dockerized GitHub Actions CI		
LLM Eval Harness	Provenance-Focused NLP Benchmark	2025
Built vendor-neutral benchmark comparing Claude, GPT-4o, and Gemini across grounded QA, multi-hop QA, and claim verification on a 2,000-example frozen set; tracked TokenF1, evidence-quote validity, and grounded hallucination rate with paired bootstrap CIs, McNemar test, and Holm correction		

Languages and Technologies

Languages: Java, Python, TypeScript, JavaScript, SQL, Bash Search & NLP: Elasticsearch, BM25, dense/cross-encoder re-ranking, sentence-transformers, PyTorch, Qdrant, RAG, NDCG/MRR evaluation Backend & Infra: Spring Boot, FastAPI, Node.js, Kubernetes, Docker, Kafka, Redis, PostgreSQL, AWS, GCP Tooling & Observability: Testcontainers, GitHub Actions, OpenTelemetry, Prometheus, Grafana		
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Additional Experience and Awards

IEEE Published: Computer Vision-Based Cybersecurity Threat Detection, Springer icSoftComp 2023; presented to 500+ attendees Hackathon Winner & GDSC Lead: Top 10 in 6+ national hackathons (3 MVPs in 48-hour sprints); led 4+ tech events reaching 500+ students; writing at kulkarniaditya.com/blog		
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